

Curriculum Vitae¹

Keisuke Hoshino

Contact details

Division of Mathematics and Mathematical Sciences, Faculty of Science, Kyoto University,
Kitashirakawa Oiwake-chō, Sakyo-ku, Kyoto-shi, Kyoto 606-8502 Japan

Email 1: hosiksk215@gmail.com

Email 2: hoshinok@kurims.kyoto-u.ac.jp

Phone: +81 80-5456-5530

Fundings

2023–2026 JSPS Research Fellowship for Young Scientists (DC1)

Education

2023– Ph.D. course in Division of Mathematics and Mathematical Sciences, Faculty of Science, Kyoto University.

Expected completion : March 2026

Thesis title : *A Homotopy Theory of Algebraic Higher Categories*

Supervisor : Masahito Hasegawa

2021–2023 Master's course in Division of Mathematics and Mathematical Sciences, Faculty of Science, Kyoto University.

Thesis title : *Towards structures of higher dimensional categories*

Supervisor : Masahito Hasegawa

2017–2021 Undergraduate course in Faculty of Science, Kyoto University.

Publications

- [1] T. Sanada, K. Hoshino, K. Hirai, and S.-y. Katsumata. “Programming Backpropagation with Reverse Handlers for Arrows”. In: *Proc. ACM Program. Lang.* 10.ICFP, 306 (Aug. 2026). 29 pages. DOI: 10.1145/3828704. URL: <https://doi.org/10.1145/3828704>.
- [2] S. Fujii, K. Hoshino, and Y. Maehara. “ ω -weak equivalences between weak ω -categories”. In: *Adv. Math.* 480 (2025), Paper No. 110490. ISSN: 0001-8708,1090-2082. DOI: 10.1016/j.aim.2025.110490. URL: <https://doi.org/10.1016/j.aim.2025.110490>.
- [3] K. Hoshino and H. Nasu. “Double categories of relations relative to factorisation systems”. In: *Appl. Categ. Structures* 33.2 (2025), Paper No. 11, 70. ISSN: 0927-2852,1572-9095. DOI: 10.1007/s10485-025-09799-y. URL: <https://doi.org/10.1007/s10485-025-09799-y>.
- [4] 眞田 嵩大, 平井 謙信, 星野 恵佑, and 勝股 審也. “逆微分圏に基づいたアローハンドラによるニューラルネットワークプログラミング言語に向けて. Towards a Neural Network Programming Language with Arrow Handlers Based on Reverse Differential Categories”. In: **日本ソフトウェア科学会第 42 回大会 (2025 年度) 講演論文集**. 査読なし論文 (This is an unrefereed paper) . 日本ソフトウェア科学会. 2025. URL: <https://jssst.or.jp/files/user/taikai/2025/papers/1a-1-R.pdf>.
- [5] S. Fujii, K. Hoshino, and Y. Maehara. “Weakly invertible cells in a weak ω -category”. In: *High. Struct.* 8.2 (2024), pp. 386–415. ISSN: 2209-0606. DOI: 10.21136/HS.2024.14.

Preprints

¹Last updated: September 2, 2026

- [1] S. Fujii, K. Hoshino, and Y. Maehara. *ω -equifibrations between strict and weak ω -categories*. 2025. arXiv: 2511.09849 [math.CT]. URL: <https://arxiv.org/abs/2511.09849>.

Talks

(Invited)

- [1] K. Hoshino and H. Nasu. *Double categories of relations relative to factorisation systems*. Second Virtual Workshop on Double Categories, Online. Oct. 2024.

(non-refereed)

- [1] K. Hoshino. *Strong subgenericity and the Structure-Semantics adjunction for T -operads*. Computer Science and Category Theory (CSCAT2026), Fukui, Japan. Mar. 2026.
- [2] K. Hoshino. 代数的入射性と *cellular replacement comonad*. Categories in Tokyo, National Institute of Informatics, Tokyo, Japan. Feb. 2026.
- [3] K. Hoshino. *Bivirtual double categories and localisation of composing cells*. Computer Science and Category Theory (CSCAT2025), Kumamoto, Japan. Mar. 2025.
- [4] K. Hoshino. *Factorisation systems and (virtual) equipments*. Computer Science Theory Seminar, Logic and Semantics Group in TTÜ, Tallinn, Estonia. Dec. 2024.
- [5] K. Hoshino and H. Nasu. *Double categories of relations relative to factorisation systems*. 67th Annual Meeting of the Australian Mathematical Society (AustMS 2023), Brisbane, Australia. Dec. 2023.
- [6] K. Hoshino. *Factorisation systems and equipments*. Australian Category Seminar, Macquarie University, Sydney, Australia. Nov. 2023.
- [7] K. Hoshino. *Towards Structures of Higher Dimensional Categorical Structures*. Computer Science and Category Theory (CSCAT2023), Kyoto, Japan. Mar. 2023.
- [8] K. Hoshino. *PCAs as algebras over operads*. Computer Science and Category Theory (CSCAT2022), Online. Mar. 2022.

Teaching experiences

- 2021 Teaching assistant in Faculty of Science, Kyoto University. Tutor for KTGU Faculty Seminar (KTGU 学部セミナー) reading *Introduction to the Theory of Computation* by Sipser, Michael.
- 2022 Teaching assistant in Faculty of Science, Kyoto University. Tutor for KTGU Faculty Seminar (KTGU 学部セミナー) reading プログラム意味論 (*Program Semantics*) by Yokouchi, Hirofumi.